

Usman Ali

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Education

Bachelors in Science, Software Engineering | Expected 2027 *COMSATS Institute of Information Technology*
CGPA: 3.12 / 4.0

Summary

Software Engineering student with a robust foundation in Python and a specialized focus on **Machine Learning**. Passionate about applying theoretical ML knowledge to develop real-world predictive models and automated systems. Experienced in building end-to-end applications from data signal processing to responsive web frontends, with a commitment to mastering industry best practices and innovative AI-driven projects.

Core Skills

- **Language:** Python, JavaScript, PHP
- **Libraries:** NumPy, Pandas, Scikit-learn, Beautiful Soup
- **Full-Stack Development:** Flask, Laravel, MERN Stack (MongoDB, Express, React, Node.js)

Projects

Image Classification System

- Developed a full-stack web application that classifies images of five historical figures with **85%+ accuracy** using Python, Flask, and Scikit-Learn.
- Engineered a custom data pipeline using **OpenCV** for facial feature extraction and **Wavelet Transform** for image signal processing to enhance model feature sets.
- Optimized model performance by performing hyperparameter tuning via **GridSearchCV** across SVM, Random Forest, and Logistic Regression models.
- Designed a responsive frontend using **Bootstrap** and **Dropzone.js**, enabling real-time image processing through asynchronous **jQuery** API calls to the Flask backend.

Project link:

<https://github.com/usman3261/Image-Classification>